

Fig. 14: Actual-size PCB layout of 7-segment display unit (Fig. 6)

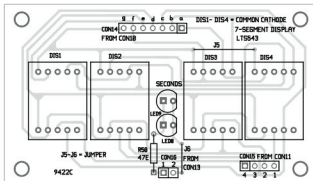


Fig. 15: Components layout for the PCB in Fig. 14

alarm values. Thereafter, it reverts to normal time display with an acknowledgement beep.

Similarly, the Minus button can be used to decrement the values in 'Settings' mode. Pressing it once during normal time display mode first displays the current date on the left and the current month on the right, and then the current year. Thereafter, it reverts to normal time display with an acknowledgement beep.

To switch on/off the alarm, simply press and hold the Minus button for more than one second. If the display indicates AL-0, the alarm is turned off. AL-1 indicates the alarm is turned on.

To set the various parameters of the clock, press SET button once to

1. Set current hours
2. Set current minutes: After setting the current hours and minutes, SET is displayed on the display
3. Set month
4. Set date
5. Set year. After setting month, date and year SET is displayed on the display
6. Set alarm hours
7. Set alarm minutes. After setting alarm hours and minutes SET is displayed on the display

To switch off the running alarm, press any of the three buttons once. Green colour on the RGB1 LED indicates AM, blue indicates PM and red indicates that date is displayed.

Construction and testing

An actual-size PCB layout of the power unit (in Fig. 3) is shown in Fig. 8 and its components layout in Fig. 9. After

bring it to 'Settings' menu. In 'Settings' menu, use Plus and Minus keys to increment and decrement the current values, respectively. The corresponding pair of 7-segment displays starts blinking. Pressing Set button again brings the clock into the next parameter setting mode and so on until the normal time is displayed. On pressing Set button once, the following seven settings are displayed:

1. Set current hours
2. Set current minutes: After setting the current hours and minutes, SET is displayed on the display

EFY Note
The source code of this project and calculations, charging.doc file are included in this month's EFY DVD and are also available for free download at source.efymag.com

assembling the circuit on the PCB, connect CON1 to the secondary of transformer X1, and CON2 to 10-12V, 7W solar panel. Extend CON3 through CON5 to other sections using proper connectors. Enclose the PCB in a suitable cabinet.

An actual-size PCB layout of the power LED and USB charging circuit (in Fig. 4) is shown in Fig. 10 and its components layout in Fig. 11. After assembling the circuit on the PCB, connect CON6 to CON3 in Fig. 3, and CON7 to CON4 in Fig. 3. Use CON8 for mobile charging through the USB cable. Affix all the LEDs (LED2 through LED7) on the top or front side of the cabinet.

An actual-size PCB layout for the main clock controller circuit (in Fig. 5) is shown in Fig. 12 and its components layout in Fig. 13. After assembling the circuit on the PCB, connect CON9 to 5.2V from CON5 in Fig. 3. Connect the RGB1 LED and switches S1 through S6 on the front side of the cabinet.

An actual-size PCB layout of the 7-segment display unit (Fig. 6) is shown in Fig. 14 and its components layout in Fig. 15. After assembling the circuit on the PCB, connect CON14 to CON10, CON16 to CON13, and CON15 to CON11 from Fig. 5. Affix LED8 and LED9 on the PCB for seconds indication. **EFY**



Abhishek Kumar is B.Tech in electronics and instrumentation. His interests include embedded systems, software programming and instrumentation

For More
Test and Equipment Stories
Visit www.electronicstest2b.com